

PANKAJ TYAGI

Full-Stack Developer | MERN • Next.js • Python CV/AI

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SUMMARY

Full-Stack Developer who ships, not just codes. Built and deployed 4 production systems end-to-end — spanning real-time CV pipelines, RAG-powered AI apps, and full e-commerce platforms — using React/Next.js, Node.js, Python, PostgreSQL, MongoDB, and Redis. Comfortable owning a feature from schema design to Vercel deployment. Looking to join a startup or product team where moving fast and taking ownership are defaults.

TECHNICAL SKILLS

Frontend: React.js, Next.js 14/15 (App Router), TypeScript, Tailwind CSS, shadcn/ui, Framer Motion, Aceternity UI

Backend: Node.js, Express.js, FastAPI, Flask, Python (Threading, Queue), REST APIs, Redis Pub/Sub, NextAuth.js

Databases: PostgreSQL, MongoDB Atlas, Redis (Upstash), Prisma ORM, Convex

CV & AI: OpenCV, face_recognition, dlib HOG, MJPEG Streaming, 128D face embeddings, RAG, Vercel AI SDK, OpenRouter API

DevOps & Tools: Docker, Vercel, Git, GitHub Actions, Cloudinary, Nodemailer, Postman, ESLint, Husky

Languages: JavaScript (ES6+), TypeScript, Python, Java

PROJECTS

- Anjani Beauties — Freelance Client Project** | Stack: *Next.js 14, TypeScript, Prisma, PostgreSQL, Nodemailer, Docker, Vercel*
 - ▶ **Delivered end-to-end** a production website for a real paying client (beauty & bridal salon) — scoped, built, and deployed solo; site is live and actively serving customers.
 - ▶ **Engineered a full inquiry pipeline** — contact and bridal-inquiry API routes with rate limiting, server-side validation, and Nodemailer email dispatch, eliminating manual booking calls for the client.
 - ▶ **Reduced booking friction** by building a multi-step bridal inquiry form backed by Prisma + PostgreSQL (Supabase), capturing structured lead data with zero manual data entry on the client's end.
 - ▶ **Shipped with production-grade tooling:** Dockerfile for Node 20 LTS, GitHub Actions CI, Husky pre-commit hooks (ESLint + Prettier + TypeScript checks), and Framer Motion animations — all on a freelance timeline.
- CV Security — Real-Time Face Recognition Access Control** | Stack: *Next.js 15, Python, OpenCV, dlib, Flask, MongoDB, Redis* | [Source Code](#)
 - ▶ **Achieved ~30 fps real-time face detection** across multiple simultaneous camera feeds using a Python producer-consumer architecture (threading + Queue); eliminated pipeline restarts on enrollment changes via Redis Pub/Sub hot-reload.
 - ▶ **Solved a hard concurrency bug:** dlib's Windows thread-safety crash under multi-camera load — fixed with a shared threading.Lock serializing all dlib calls, achieving zero-crash stability in production testing.
 - ▶ **Streamed live annotated video** (OpenCV bounding boxes + labels) to a Next.js dashboard via Flask MJPEG server using a single tag — no WebRTC complexity, zero extra dependencies on the frontend.
 - ▶ **Shipped role-based access** (SUPER_ADMIN / ADMIN) with NextAuth.js JWT auth and a paginated event log backed by MongoDB + Cloudinary face-crop uploads — fully deployed on Vercel.
- Brain Notes AI — RAG-Powered Notes App** | Stack: *Next.js, Convex, Vercel AI SDK v5, OpenRouter API* | [Source Code](#)
 - ▶ **Built a context-aware AI chatbot** that answers questions using the user's own notes as a retrieval corpus (RAG) — grounding responses in personal knowledge rather than generic LLM output.
 - ▶ **Implemented real-time tool calling** with Vercel AI SDK v5 and Convex backend. Enabling live data retrieval within a single chat turn with secure auth and low-latency response.

4. **Fin Morph AI — Finance Management Platform** | Stack: *Next.js, Clerk, PostgreSQL, OCR* | [Source Code](#)

- **Cut manual data entry by 50%** by building a receipt OCR + automated parsing pipeline that categorises and logs transactions without user input.
- **Automated monthly budget reports** with data-driven insights and exportable financial summaries — delivered via a dashboard with spending trend visualisations and Clerk-secured sessions.

EXPERIENCE

1. **Web Developer Intern | EY Technology (Virtual)** *Feb 2024 – Apr 2024 • Remote • 6 weeks*
 - **Improved user engagement metrics** on a Django music streaming platform by implementing performance fixes and overhauling EJS-based navigation UI — changes shipped to a production codebase.
 - **Delivered feature work and bug fixes** across database management and authentication systems, gaining hands-on experience in enterprise-grade Python backends
2. **AI & Cloud Intern | Edunet Foundation (Virtual)** *Feb 2024 – Mar 2024 • Remote • 4 weeks*
 - **Built and deployed an NLP chatbot** using IBM Watson; applied conversational AI design patterns and cloud deployment strategies within a structured 4-week programme

EDUCATION

B.Tech — Computer Science Engineering *2021 – 2025*
Doon Institute of Engineering & Technology, Rishikesh (Uttarakhand Technical University)

ACHIEVEMENTS & OPEN SOURCE

- **Open-Source Contributor Open-Source Contributor** — feature enhancements and bug fixes in the React and JavaScript ecosystems.
- **Certified: MERN Full-Stack Development** — DevRhyme Foundation (2024), awarded after practical evaluation.
- **Hackathon Participant** — designed innovative project in Unstop Hackathon 2023 (competitive team environment)